

Continued Fractions

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1 Convergence

Let $\{u_n\}, \{l_n\}$ be decreasing and increasing sequences, resp. s.t. $u_n \geq l_n$ for all n . Define

$$k_n = \begin{cases} u((n+1)/2) & \text{if } n \text{ is odd,} \\ l(n/2) & \text{if } n \text{ is even.} \end{cases}$$

Then $\{k_n\}$ converges iff $(u_n - l_n) \rightarrow 0$.

If $\{k_n\}$ converges, then so do $\{u_n\}, \{l_n\}$, and to the same limit L . Then $(u_n - l_n) \rightarrow L - L = 0$. Sps FTSOC $l_n > u_m$ for some n, m . Then $l_n > u_m \geq l_m$, so $n > m$. Now $l_n \leq u_n \leq u_m < l_n$, cont. Hence, $l_n \leq u_m$ for all n, m , so $\{u_n\}, \{l_n\}$ are bounded below and above, resp, and therefore converge, say to L, M . Since $(u_n - l_n) \rightarrow L - M = 0$, $L = M$, so $\{k_n\}$ converges.

2 Square Roots

Let

$$\frac{\sqrt{Q} + a_n}{b_{n+1}} = c_{n+1} + \frac{\sqrt{Q} - c_{n+1}b_{n+1} + a_n}{b_{n+1}} = c_{n+1} + \frac{Q - (c_{n+1}b_{n+1} - a_n)^2}{b_{n+1}(\sqrt{Q} + (c_{n+1}b_{n+1} - a_n))},$$

so $a_{n+1} = c_{n+1}b_{n+1} - a_n$ and $b_{n+2} = \frac{Q - (c_{n+1}b_{n+1} - a_n)^2}{b_{n+1}} = \frac{Q - a_{n+1}^2}{b_{n+1}}$. Also, take c_{n+1} to be maximal, so $c_n \neq 0$ always (for $\sqrt{Q} \notin \mathbb{Z}$). Let $a_0 = 0$ and $b_1 = 1$. Note $Q - a_{n+1}^2 \equiv Q - a_n^2 \equiv 0 \pmod{b_{n+1}}$, so $b_n \in \mathbb{Z}$ by induction.

Note $0 < b_{n+2} \leq Q$ and $c_n^2 \geq c_n$, so from $Q = a_{n+1}^2 + b_{n+1}b_{n+2} \geq a_{n+1} + b_{n+2}$, there are finitely many choices of a_n, b_n , so $\{a_n\}, \{b_n\}$ are periodic.

$1 + c_{n+1} > \frac{\sqrt{Q} + a_n}{b_{n+1}}$, so $a_{n+1} + b_{n+1} > \sqrt{Q}$.